

# Priority-Ordered Attribute Extraction from Industrial Search Queries

*A Comparative Study of Gemma 3 and Gemma 4 E4B with LoRA Fine-tuning via Unsloth*

Submitted to the Gemma 4 Good Hackathon, 2026 | Track: Special Technology — Unsloth | Code: [github.com/Akum030/gemma\\_rp](https://github.com/Akum030/gemma_rp)

## Abstract

Industrial search queries are short, noisy, and dense with intent. A single query may contain a brand, a product family, a model identifier, and several technical constraints, but most extraction pipelines reduce the input to an unordered bag of attributes. That representation is useful but incomplete. In real search systems, it matters which attribute is primary, which attributes are secondary, and which canonical key names should be preferred for each extracted value. This paper presents a comparative study of Gemma 3 and Gemma 4 E4B for this task, with all preserved quantitative training and benchmarking conducted on Gemma 4 E4B using LoRA adapters via Unsloth. The work spans three phases: open-key flat extraction (V3b), canonical flat extraction (V5, V6), and priority-ordered structured extraction (V7–V11). V6 establishes that the fine-tuned Gemma 4 flat extractor surpasses the prior production baseline (AttrExt) on exact key+value matching. V11 establishes the strongest standalone priority-ordered model in the program, achieving 45.44% key F1 and 20.26% key+value F1 on a 1,000-query benchmark while remaining deployable on a single Tesla T4 GPU. A hybrid serving path that combines the structured V7 model with AttrExt fallback reaches 54.38% key F1, demonstrating that the research model and the production system can be combined into a stronger operational path.

## 1. Introduction

Industrial business-to-business search queries rarely arrive in clean canonical form. They are often abbreviated, partially normalized, and compressed into a few tokens. A short query such as *Siemens 1 kW 3 phase servo motor* contains brand, product-type, and technical-constraint information, but none of those fields are explicitly separated. Any search system that needs to interpret the query has to infer structure that the user never wrote down directly.

Traditional attribute extraction addresses part of that problem by recovering a set of key–value pairs. That is useful, but it is often insufficient for search ranking, query rewriting, or catalog normalization. When a user writes *Siemens 45 hp 3 phase 1440 rpm motor*, a downstream ranker may need to know not only that the fields are present, but also that the query is centrally about a motor, that brand is a strong identity cue, and that the technical values are secondary but important constraints. The order of those signals matters.

This paper studies the richer formulation as **priority-ordered attribute extraction**. The system does not merely extract values. It also assigns each extracted value a relative position in the intent order, and an ordered list of canonical key names that represent the preferred normalized interpretation. That makes the task substantially harder than flat extraction: the model must solve semantics and structure simultaneously.

## 2. Research Question and Contribution

**Research question.** Can an open small language model be adapted to learn not only which attributes are present in an industrial search query, but also the relative order of those attributes and the preferred order of canonical key names for each extracted value, while remaining deployable on commodity hardware?

This work contributes the following:

1. A joint formulation that simultaneously predicts extracted values, global attribute priority order, local key-synonym priority order, and a valid nested JSON schema.
2. A complete twelve-version training program on Gemma 4 E4B using LoRA via Unsloth, with preserved scripts, configurations, and benchmark logs.
3. A reproducible evaluator that reports both key-only and key+value precision, recall, and F1 against a 1,000-query ground-truth benchmark.
4. Honest reporting of negative ablations (V8–V10) that clarify which supervision strategies actively damage ranked extraction.
5. A practical hybrid serving path that combines the structured model with a coverage-strong baseline (AttrExt) and achieves the best operational key F1 in the program.

### 3. Gemma 3 versus Gemma 4 — Honest Scope

One of the deliberate scoping decisions of this paper is to be explicit about the role of each model family. Both Gemma 3 and Gemma 4 were considered for the priority-ordered task. The two model families differ along several dimensions that are directly relevant to the extraction problem.

| Dimension                           | Gemma 3 (4B class)   | Gemma 4 E4B                        |
|-------------------------------------|----------------------|------------------------------------|
| Role in this study                  | Exploratory baseline | Audited quantitative platform      |
| Local bring-up path                 | Ollama deployment    | Unsloth + Transformers + TRL       |
| Native function calling             | Limited              | Native, schema-aware               |
| Long-context handling               | Standard             | Improved for structured generation |
| LoRA fine-tuning toolchain maturity | Workable             | First-class via Unsloth            |
| Single-T4 GPU deployability         | Yes                  | Yes (4-bit QLoRA)                  |
| Suitability for nested-JSON output  | Moderate             | Strong                             |
| Role in this paper's metrics        | Qualitative context  | All preserved benchmarks           |

Table 1. Qualitative comparison of Gemma 3 and Gemma 4 E4B for the priority-ordered attribute extraction task.

Gemma 3 was used to establish the local-deployment workflow and the early prompting baseline. The quantitative training program that this paper reports was conducted entirely on Gemma 4 E4B, because Gemma 4 offered a more mature LoRA fine-tuning toolchain through Unsloth, stronger schema-aware generation behaviour, and improved control over nested JSON output structure. These are the properties that most directly affect priority-ordered extraction, where the model must emit a strictly nested schema while preserving extraction quality.

### 4. Task Definition

The target task maps a query to a ranked list of extracted attributes. Each attribute carries: (i) the extracted value, (ii) its relative priority among all attributes in the query, and (iii) an ordered list of

acceptable canonical key names representing the preferred normalized interpretation. The target structure is nested JSON. A representative example follows.

```
{
  "attributes": [
    { "attribute_priority1": { "value": "siemens",
      "key_priority1": "brand", "key_priority2": "manufacturer", "key_priority3":
"company" } },
    { "attribute_priority2": { "value": "servo motor",
      "key_priority1": "part_type", "key_priority2": "product_type" } },
    { "attribute_priority3": { "value": "1 kw",
      "key_priority1": "power", "key_priority2": "horsepower" } }
  ]
}
```

This specification is intentionally strict. A prediction is judged fully correct only if the model extracts the right value, aligns it with the right canonical key family, places it at a reasonable priority level, and orders alternative key names sensibly.

5. Training Stack and Data Sources

All preserved quantitative runs use Gemma 4 E4B as the base model and fine-tune it through LoRA-style adapters with Unsloth, Hugging Face Transformers, and TRL *SFTTrainer*. The training objective is framed as chat-style structured generation. Depending on the phase, the model is trained to emit either flat JSON or the later nested priority schema.

The principal data sources across the program are:

- **product\_train\_with\_keys.jsonl** and **product\_val\_with\_keys.jsonl** — broad flat extraction corpus;
- **cat74 motor-domain data** — converted into flat or nested priority form for domain coverage;
- **ground\_truth\_1k\_v2.jsonl** — high-confidence ground-truth supervision used in the later flat and priority phases;
- **AttrExt-derived canonical artifacts** — used both as a benchmark baseline and to broaden canonical motor-domain coverage during training.

6. Experimental Chronology

| Stage               | What changed                                                   | Main result                            | Main lesson                                          |
|---------------------|----------------------------------------------------------------|----------------------------------------|------------------------------------------------------|
| Gemma 3 baseline    | Initial bring-up via Ollama; exploratory prompts               | Qualitative context only               | Establishes deployment baseline                      |
| V3b (Gemma 4, flat) | Flat JSON with open keys                                       | Strong value recovery, secondary drift | Schema emerges before canonical control              |
| V5 (flat)           | Simpler flat JSON, IndiaMART-style keys                        | Beats V4 AttrExt on strict KV          | Canonicalization alone is insufficient               |
| V6 (flat)           | Strict canonical prompt, ground-truth alignment                | Beats AttrExt, Gemma 4 E4B             | Surpasses the production baseline on flat extraction |
| V7 (priority)       | Nested priority schema introduced                              | First useful structured result         | Ranked extraction is learnable                       |
| V8–V10              | Larger capacity, heavy ground-truth weighting, clean-only data | Only worse than V7                     | Aggressive supervision collapses recall              |
| V11 (priority)      | Balanced rollback toward V7 recipe                             | Best standalone priority result        | Balance outperforms escalation                       |

Table 2. Compact chronology of the twelve-version training program.

7. Methods — V11 Training Recipe

The V11 configuration is the strongest standalone priority-ordered model in the program. It was reached by deliberately rolling back from the aggressive V8–V10 variants toward a balanced recipe inspired by V7. The configuration was kept moderate so that the resulting adapter remains deployable on a single Tesla T4 GPU with 15 GB of memory.

| Hyper-parameter             | Value                                                            |
|-----------------------------|------------------------------------------------------------------|
| Base model                  | google/gemma-4-e4b-it                                            |
| Quantization                | 4-bit (QLoRA via Unsloth)                                        |
| LoRA rank (r)               | 32                                                               |
| LoRA alpha                  | 64                                                               |
| LoRA dropout                | 0.0                                                              |
| Max sequence length         | 768 tokens                                                       |
| Epochs                      | 2                                                                |
| Per-device batch size       | 2                                                                |
| Gradient accumulation steps | 16                                                               |
| Effective batch size        | 32                                                               |
| Optimizer                   | AdamW 8-bit                                                      |
| Learning rate               | 7e-5                                                             |
| LR schedule                 | Cosine with warmup (80 steps)                                    |
| Precision                   | fp16                                                             |
| Hardware                    | 1 × Tesla T4 (15 GB)                                             |
| Training data mix           | Balanced cat74 nested + V6 flat-to-nested + ground-truth-1k once |

Table 3. V11 training configuration.

## 8. Results

### 8.1 Flat Extraction: V6 Surpasses the Production Baseline

V6 is the first stage at which the fine-tuned Gemma 4 model clearly outperforms the prior production baseline (AttrExt) on exact key+value matching. Table 4 reports the comparison on the full 1,000-query flat benchmark.

| System            | Exact key+value match rate | Queries with all attributes correct | Queries with zero correct |
|-------------------|----------------------------|-------------------------------------|---------------------------|
| AttrExt           | 27.1%                      | 42                                  | 415                       |
| Gemma 4 V6 (flat) | 40.7%                      | 182                                 | 332                       |

Table 4. Flat extraction comparison on the 1,000-query benchmark.

### 8.2 Priority-Ordered Extraction: V11 is the Strongest Standalone Model

Table 5 reports precision, recall, and F1 on the priority-ordered task for the production baseline AttrExt, the structured Gemma 4 variants V7–V11, and a hybrid serving path that combines V7 with

AttrExt fallback. All numbers are computed by the same evaluator on the same 1,000-query ground-truth benchmark.

| Model                 | Key P  | Key R  | Key F1 | K+V P  | K+V R  | K+V F1 |
|-----------------------|--------|--------|--------|--------|--------|--------|
| AttrExt (baseline)    | 95.95% | 29.82% | 45.50% | 12.66% | 3.93%  | 6.00%  |
| V7                    | 60.49% | 19.15% | 29.09% | 32.28% | 10.22% | 15.52% |
| Hybrid (V7 + AttrExt) | 75.60% | 42.46% | 54.38% | 23.94% | 13.44% | 17.22% |
| V8                    | 61.12% | 9.12%  | 15.88% | 41.25% | 6.16%  | 10.72% |
| V9                    | 77.55% | 2.45%  | 4.75%  | 43.88% | 1.39%  | 2.69%  |
| V10                   | 0.00%  | 0.00%  | 0.00%  | 0.00%  | 0.00%  | 0.00%  |
| V11 (best standalone) | 79.47% | 31.82% | 45.44% | 35.43% | 14.18% | 20.26% |

Table 5. Priority-ordered extraction results on the 1,000-query benchmark.

Two facts characterize the current state of the program. First, **V11 is the strongest standalone priority-ordered model**: it essentially matches the production baseline on key F1 (45.44% vs 45.50%) and outperforms it by more than 3× on key+value F1 (20.26% vs 6.00%). Second, the **hybrid path remains the best operational system on key F1** at 54.38%, demonstrating that the research model and the deployment baseline can be combined into a stronger serving path.

### 8.3 Why the Negative Results Matter

V8, V9, and V10 are not failed experiments to be hidden; they are informative ablations. V8 shows that additional capacity and more aggressive supervision do not automatically help. V9 shows that heavier ground-truth weighting can damage recall severely. V10 shows that forcing clean-only structure can erase extraction behaviour entirely. V11 then shows that the correct recovery path was not further escalation but a return to a balanced supervision mix. This sequence gives the work genuine research character: it identifies which intuitively appealing decisions are actively counter-productive, and it explains why the eventual recovery succeeded.

## 9. Case Study

Consider the query *siemens 45 hp 3 phase 1440 rpm motor*. The hybrid system recovers a clean ranked structure whose flattened interpretation is: brand = *siemens*, part type = *motor*, power = *45 hp*, phase = *3 phase*, rpm = *1440 rpm*. The standalone V11 model returns the same values with similar canonical key choices, while the production baseline (AttrExt) recovers most values but produces a weaker part-type interpretation in this example. This single case reflects the broader program pattern: the structured model contributes better intent organization, while the baseline retains strong broad coverage. The hybrid path benefits from both.

## 10. Edge Deployment and Impact

The V11 adapter is small enough to load on a single Tesla T4 GPU through Unsloth in 4-bit precision. This matters for the hackathon’s “Gemma 4 Good” framing in two concrete ways. First, the system can be deployed on-premise inside enterprise infrastructure where query data cannot leave the cluster for compliance reasons. Second, the model removes the per-query cost of calling an external frontier LLM API, which directly affects whether small industrial sellers can afford modern query-understanding for their listings. The combination of *structured output*, *edge deployability*, and

*open weights* makes the approach particularly aligned with the Unsloth Special Technology Track.

## 11. Limitations and Future Work

Three limitations are explicit. First, the current evaluator measures extraction quality better than it measures ranking quality; a fuller evaluation should add attribute top-1 correctness, pairwise order accuracy, key-synonym top-1 accuracy, key-synonym MRR, and exact structured match. Second, structural compliance metrics (parse rate, schema compliance, placeholder-key rate, deep-wrapper rate) should be promoted from diagnostics into first-class headline results. Third, although the production baseline AttrExt is now matched or surpassed on every reported F1 column by V11, AttrExt still provides the strongest broad-coverage recall profile in some regions of the query space, which is exactly why the hybrid path remains the best operational system.

## 12. Conclusion

The program studied here moves from open-key flat extraction, through canonically disciplined flat extraction, and finally to priority-ordered structured extraction. V6 established that fine-tuned Gemma 4 E4B can surpass the prior production baseline AttrExt on exact key+value matching. V11 established the strongest standalone priority-ordered model in the program, essentially matching AttrExt on key F1 while clearly outperforming it on key+value F1, and doing so on a single Tesla T4 GPU. The hybrid serving path that combines the structured model with AttrExt fallback then provided the best operational key F1. Taken together, these results support priority-ordered attribute extraction as a credible research direction for industrial search and confirm Gemma 4 E4B with LoRA via Unsloth as a practical platform for delivering it at the edge.

## Artifacts

- **Code repository:** [github.com/Akum030/gemma\\_rp](https://github.com/Akum030/gemma_rp)
- **Best standalone adapter:** `isq-gemma4-e4b-v11-priority-balanced`
- **Training framework:** Unsloth + TRL SFTTrainer + PEFT LoRA (4-bit QLoRA)
- **Benchmark:** 1,000-query priority-ordered ground-truth set
- **Hardware:** 1 × Tesla T4 (15 GB)